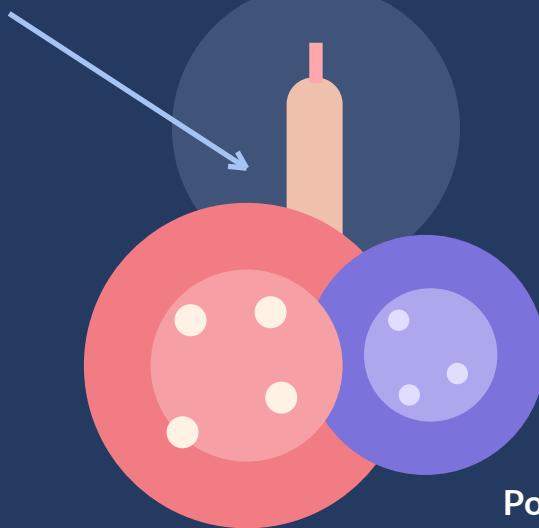


THE ANTERIOR PITUITARY

A colourful, slide-by-slide guide to the tiny gland with an enormous job.

Hypothalamus



Anterior lobe

ORAL ECTODERM

Posterior lobe

NEURAL TISSUE

WHAT IS INSIDE

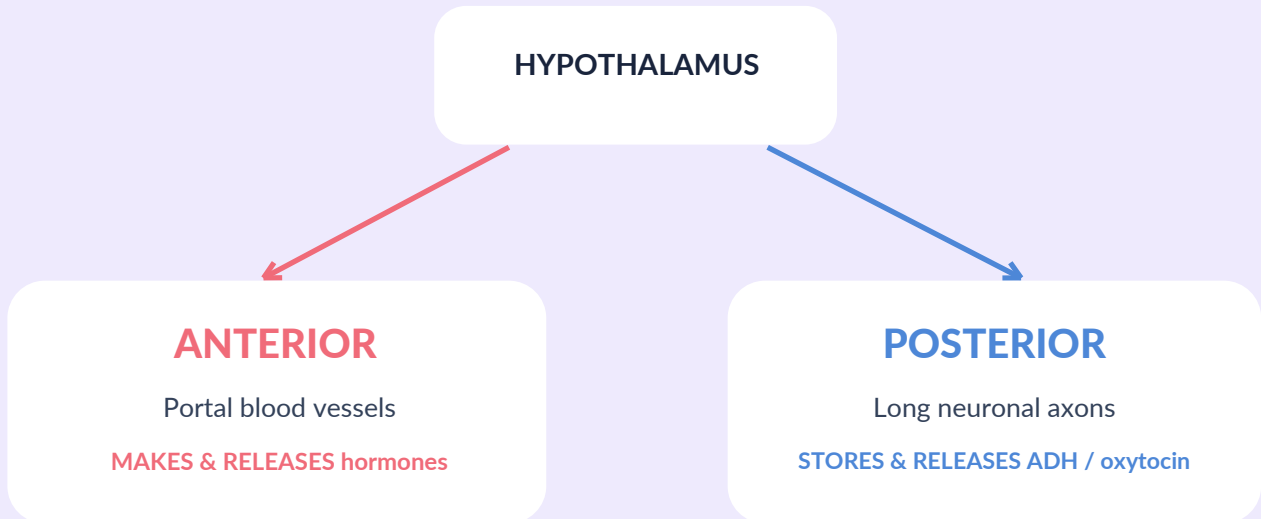
Diagrams | real lecture images | memory hooks | fun facts

Every chapter gives exact source slide numbers so you can follow your lecture.

One gland, two very different jobs

Start with this mental map: the hypothalamus runs two delivery routes.

THE BIG PICTURE



READING ROUTE

Your six checkpoints

01

Slides 4-9

Brain -> gland: anatomy & control

02

Slides 10-11

Embryology: Rathke's pouch

03

Slides 12-16

Histology: meet the 5 cell types

04

Slides 17-20

Tumours & development genes

05

Slides 21-22

Six hormones in three families

06

Slides 23-28

Prolactin, breastfeeding & disease

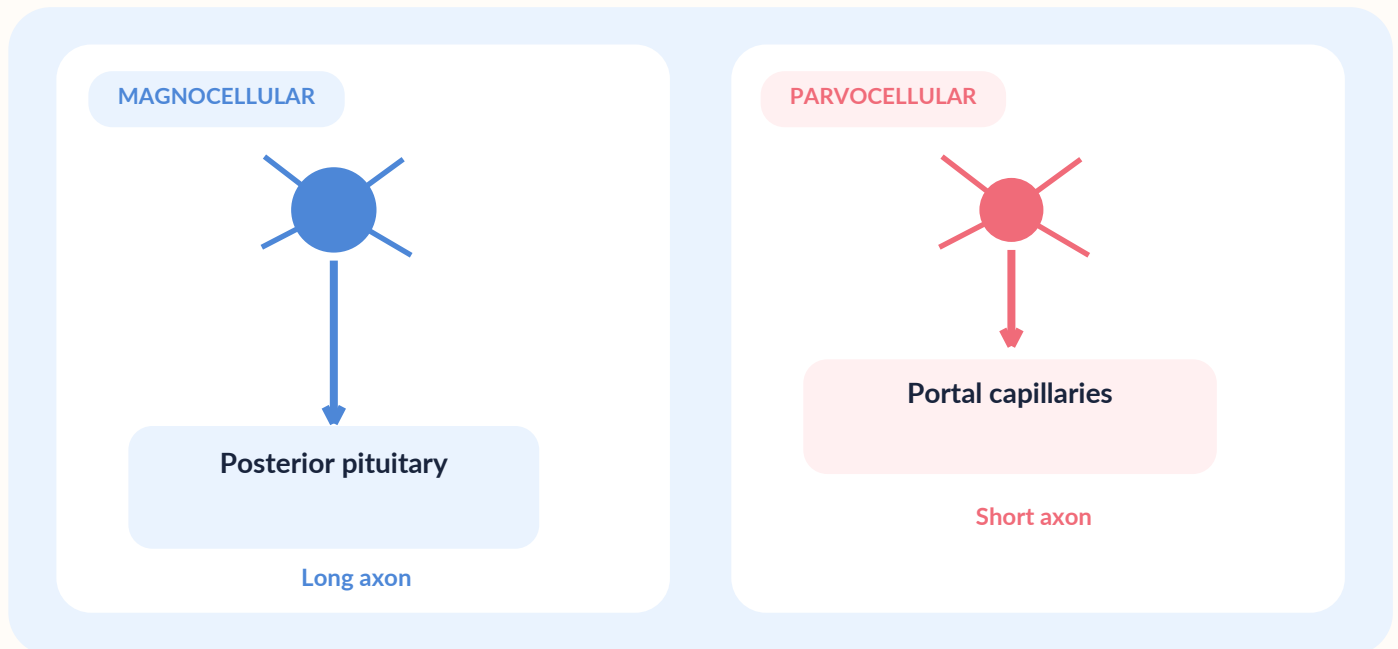


MEMORY HACK

Think: the posterior pituitary is a **delivery depot**; the anterior pituitary is a **hormone factory**.

Two types of neuroendocrine neurons

Slide 4: both start in the hypothalamus, but the routes are different.



REMEMBER THIS

Same starting point. Different destination.

● Magnocellular

Oxytocin and vasopressin (ADH) travel down long axons and enter systemic blood at the posterior pituitary.

● Parvocellular

TRH, CRH, GHRH, GnRH, somatostatin and dopamine enter portal vessels to regulate anterior pituitary cells.



WORD TRICK

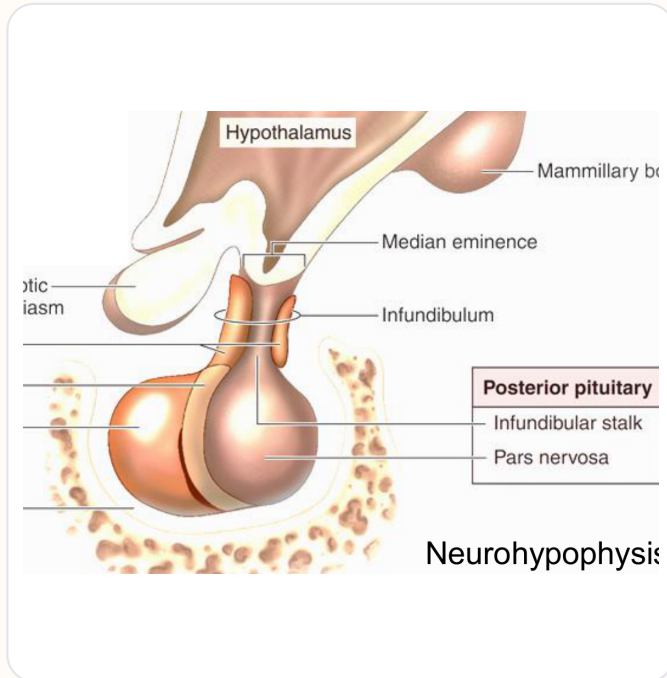
MAGNO = mega neuron + mega-long route. **PARVO** = petite neuron; its parcel goes by the **portal** system.

CHECK YOURSELF

Which route carries dopamine? Answer: the **parvocellular -> portal -> anterior pituitary** route.

Know the pituitary neighbourhood

Slides 6-7: anatomical naming + why the portal system is so efficient.



ANATOMY IN 30 SECONDS

Adenohypophysis

Anterior: pars distalis, pars tuberalis and pars intermedia.

Neurohypophysis

Posterior: pars nervosa and infundibular stalk.

Nearby: optic chiasm sits above the gland.

Lecture image: slide 6 (anatomical names).

HOW IT WORKS

The portal express lane

1 Hypothalamic regulatory hormones



2 Primary capillary plexus



3 Hypophyseal portal veins



4 Secondary plexus in anterior pituitary



WHY TWO CAPILLARY NETWORKS?

They shuttle hypothalamic hormones straight to their anterior pituitary targets, avoiding dilution in the general circulation. The local capillaries are **fenestrated** (have small openings).

Meet the five control circuits

Slide 8: learn each chain as a three-stop journey.

REPRODUCTION

GnRH



FSH + LH



Gonads

Germ cells & sex hormones

GROWTH

GHRH



GH



Liver + tissues

IGF-1, growth & metabolism

THYROID

TRH



TSH



Thyroid

Thyroid hormone production

ADRENAL

CRH



ACTH



Adrenal cortex

Cortisol production

MILK

INHIBITS!

Dopamine



Prolactin



Mammary glands

Milk synthesis

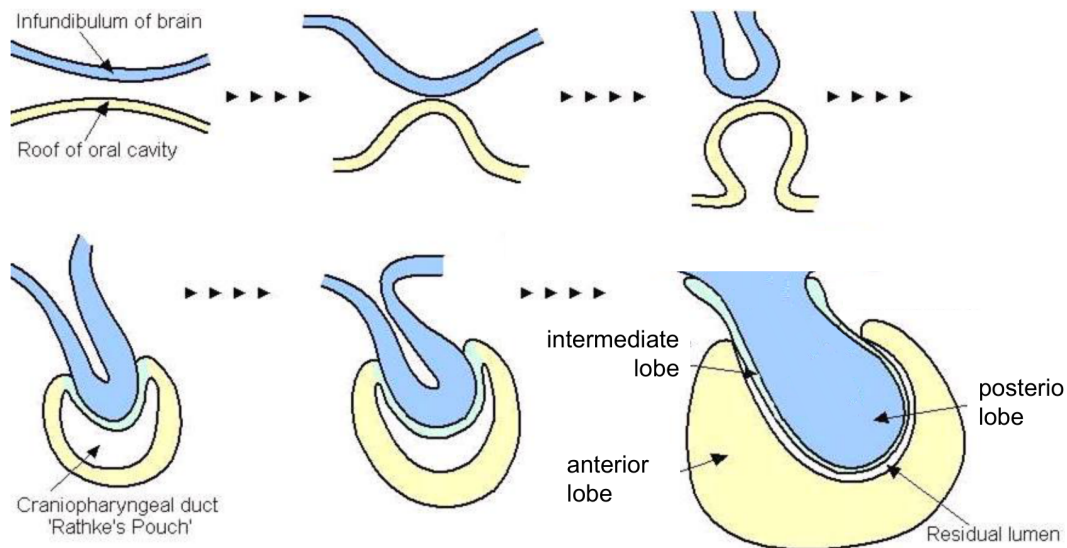


THE ODD ONE OUT

Dopamine is mainly a **brake**, not a accelerator: LESS dopamine -> MORE prolactin. Somatostatin also acts as a brake on GH.

A meeting between mouth and brain

Slides 10-11: the two lobes have different embryological origins.



Lecture developmental sequence: slide 10 (yellow = oral, blue = neural).

● Anterior = oral ectoderm

Rathke's pouch grows upwards from the roof of the developing oral cavity. It forms the glandular anterior pituitary.

● Posterior = neural tissue

The infundibulum extends downwards from the developing brain and forms the posterior pituitary.

THE INTERMEDIATE ZONE

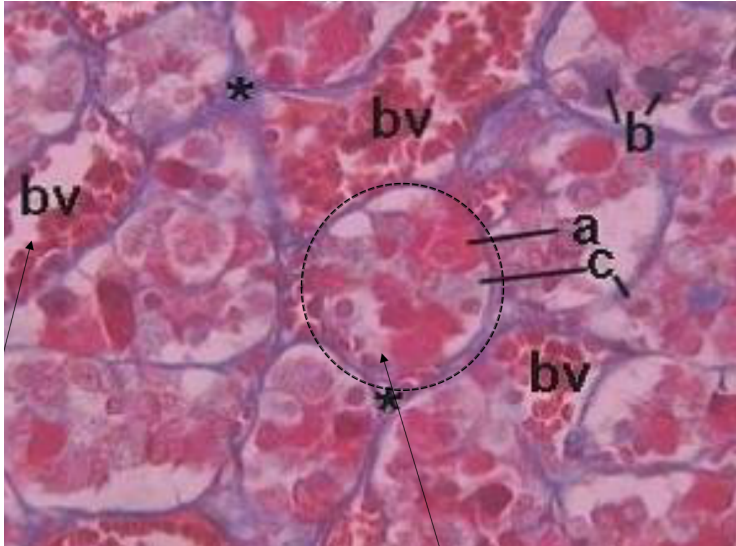
Pars intermedia sits between the anterior and posterior regions. The slide shows colloid-filled cysts - remnants of Rathke's pouch. It is much less prominent in adult humans than in many other species.

! WHY THE EMBRYOLOGY MATTERS

A craniopharyngioma can arise from embryonic epithelial remnants associated with Rathke's pouch. Developmental leftovers can become a clinical problem.

Spot the secretory cells

Slides 12-13: cell clusters sit next to capillaries so hormones can reach blood.



HOW TO READ THE IMAGE

bv = blood vessel

Endocrine cells form clusters next to blood sinuses.

A = acidophils

B = basophils

C = chromophobes

The blue stain outlines connective tissue.

Real anterior pituitary histology: slide 13.

COLOUR CODE

Three staining families

● Acidophils

Pink/red: somatotrophs (GH) and lactotrophs (prolactin).

● Basophils

Blue/purple: gonadotrophs, corticotrophs and thyrotrophs.

● Chromophobes

Pale cells that stain poorly; lecture calls them "retired" endocrine cells.

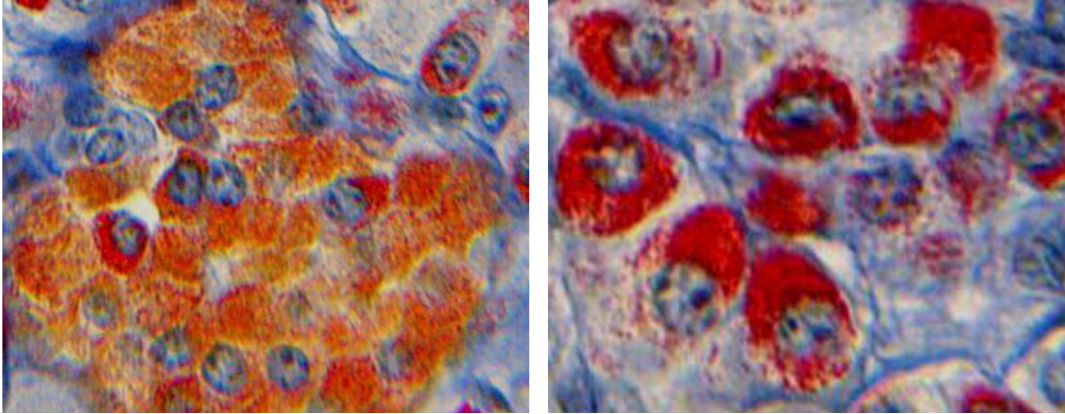


VISUAL MEMORY

Acidophils = S + L: Somatotroph + Lactotroph. **Basophils = G + C + T:** Gonadotroph + Corticotroph + Thyrotroph. Two vs three!

Five cells, six major hormones

Slides 14-16: pair every cell with its secretion before memorising morphology.



Lecture micrographs: somatotrophs (left) and lactotrophs (right), slide 15.



Somatotroph

GH

acidophil - 35-45%



Lactotroph

Prolactin

acidophil - 15-25%



Gonadotroph

FSH + LH

basophil - 10-15%



Thyrotroph

TSH

basophil - ~5%



Corticotroph

ACTH

basophil - 15-20%

WHO'S MOST COMMON?

Somatotrophs

35-45%

MICROSCOPE TIPS

Visual traits mentioned in the lecture

Somatotroph: large, clustered, large granules | **Lactotroph:** solitary, eccentric nucleus, large granules

Gonadotroph: small round, strongly basophilic | **Thyrotroph:** large, pale and angular | **Corticotroph:** pale, weakly basophilic



SPEED DRILL

Say it out loud: S -> GH; L -> PRL; G -> FSH/LH; T -> TSH; C -> ACTH.

When pituitary growth causes problems

Slides 17-18: hormone excess and pressure effects are different mechanisms.

FUNCTIONING ADENOMA

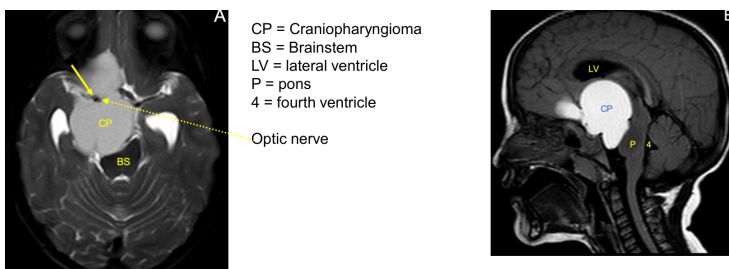
Produces excess hormone

Example: a prolactin-producing adenoma can produce high circulating prolactin.

NON-FUNCTIONING ADENOMA

May still cause compression

Pressure on normal pituitary -> hormone deficiency; pressure near optic chiasm -> visual field problems.



CRANIOPHARYNGIOMA

Usually benign, slow-growing epithelial tumour linked to Rathke's pouch remnants.

Often suprasellar: it may affect the hypothalamus, pituitary or optic chiasm.

Lecture MRI images: craniopharyngioma, slide 18.

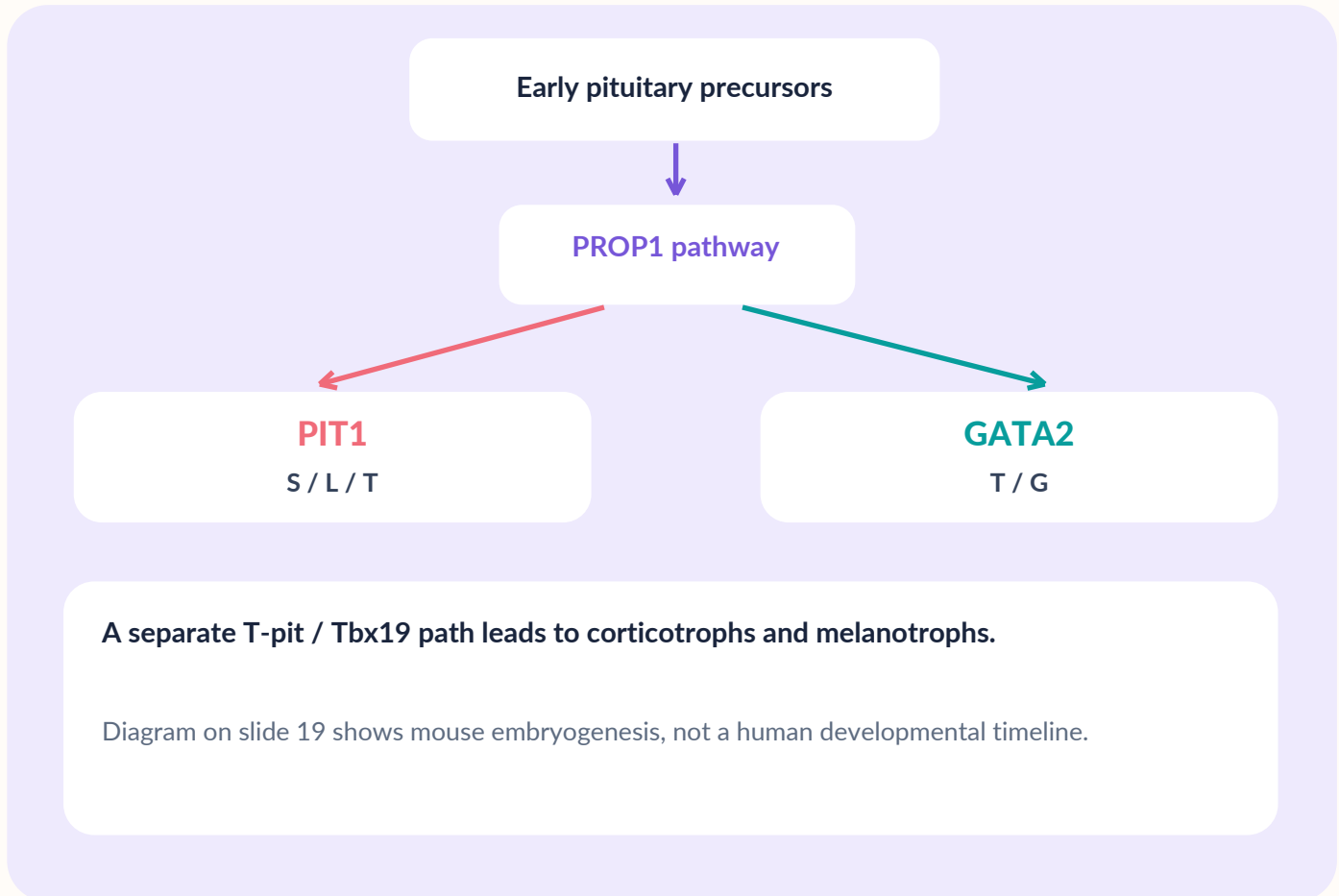


LOCATION, LOCATION, LOCATION

The pituitary sits near the **optic chiasm**. A tumour need not secrete hormones to cause symptoms; growing in a crowded neighbourhood can be enough.

Pituitary cells need a genetic script

Slides 19-20: transcription factors help developing cells choose a hormone-producing identity.



MOUSE MODELS

What happens when the script breaks?

● Ames mouse: PROP1 mutation

Reduced PRL, GH, TSH, LH and FSH;
sub-/infertility; approximately 30% of normal
size.

● Snell mouse: PIT1 mutation

Reduced PRL and GH (also TSH); similar small
phenotype.



GENE MEMORY

PIT1 looks after **three** key cell lines: Somatotroph, Lactotroph, Thyrotroph. The mouse longevity observation (~50% longer) is **mouse-model data**, not a prediction for humans.

Six hormones, three families

Slides 21-22: organise the hormones into memorable groups.

THE SIX MAJOR ANTERIOR HORMONES

ACTH

Corticotroph

LH + FSH

Gonadotroph

GH

Somatotroph

PRL

Lactotroph

TSH

Thyrotroph

FAMILY TREE

Same family does not mean same job

● Melanocortins

ACTH, lipotropin (LPH) and MSH all come from the common precursor POMC.

● Somatomammotropins

Growth hormone (GH), prolactin (PRL) and placental lactogen (PL).

● Glycoprotein hormones

LH, FSH and TSH are composed of alpha and beta subunits. They share a common alpha subunit, while the beta subunit is different.

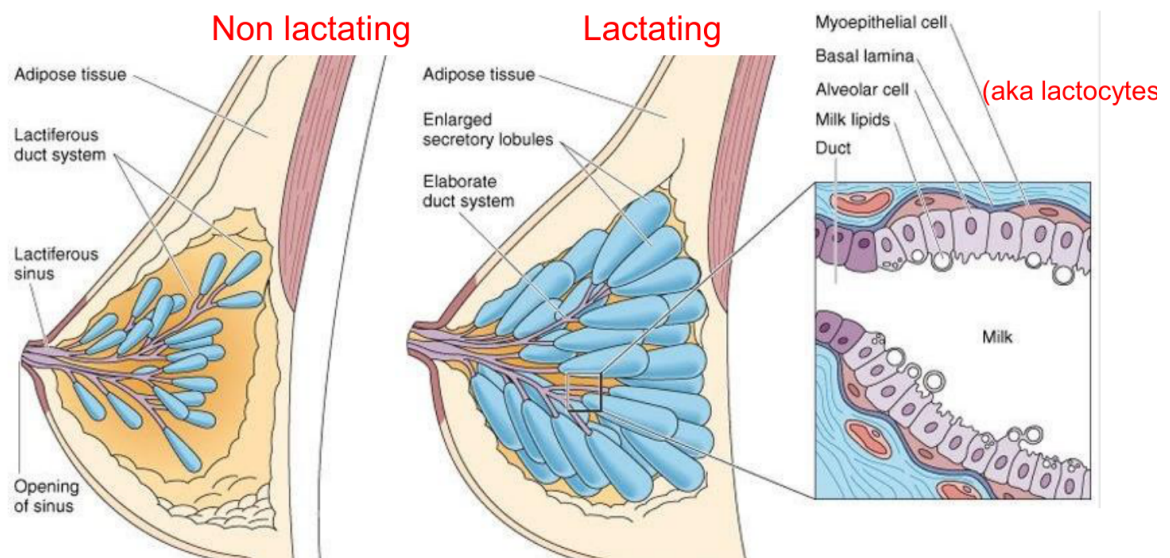
! THE LEGO TRICK

For LH / FSH / TSH, imagine a **shared alpha Lego block** plus a **different beta block** to make each hormone. Prolactin stands out because milk production is its major non-trophic action.

Prolactin: make the milk

Slides 23-25: the mammary gland develops before it can manufacture milk.

also necessary for breast milk production



Lecture diagram: non-lactating vs lactating breast, slide 24.

● MAMMOGENESIS

Development of the functional breast. Starts at puberty; increases during pregnancy with oestrogen, progesterone and prolactin.

● LACTOGENESIS

Milk production. Prolactin is necessary for production before and after birth.

FOLLOW THE ALVEOLI

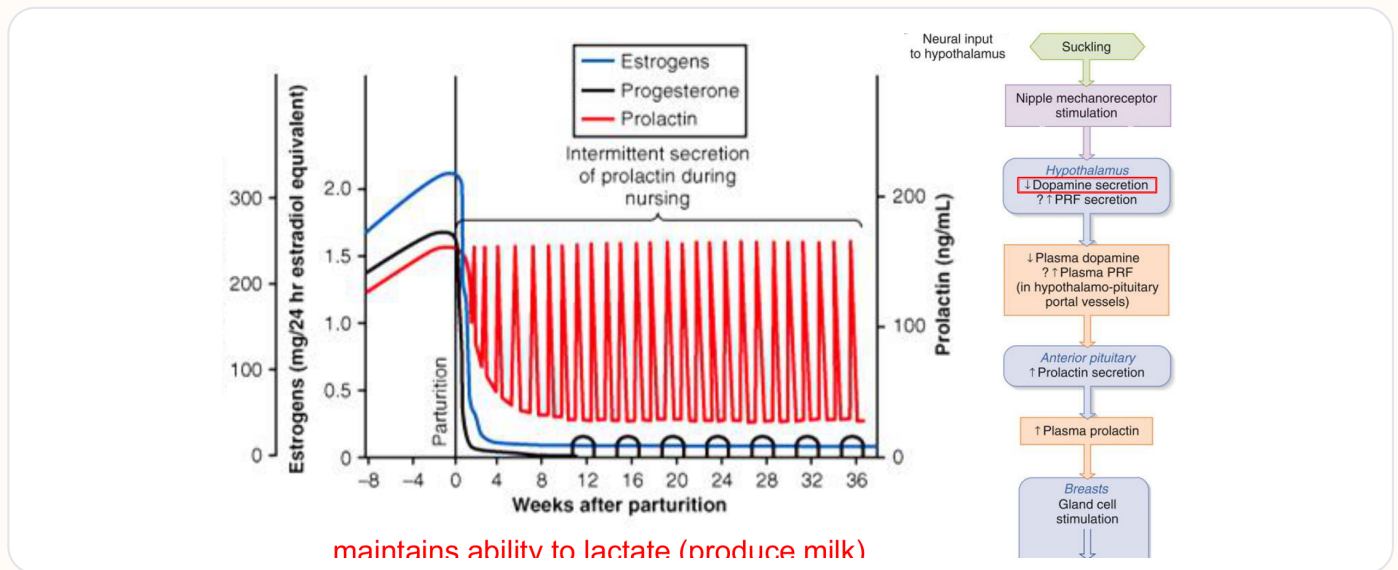
The blue lobules in the lecture image contain secretory alveoli. Their milk-producing epithelial cells are **lactocytes**. Prolactin promotes lobuloalveolar development especially in the latter half of pregnancy.

! REAL-LIFE DISTINCTION - EXTRA CONTEXT

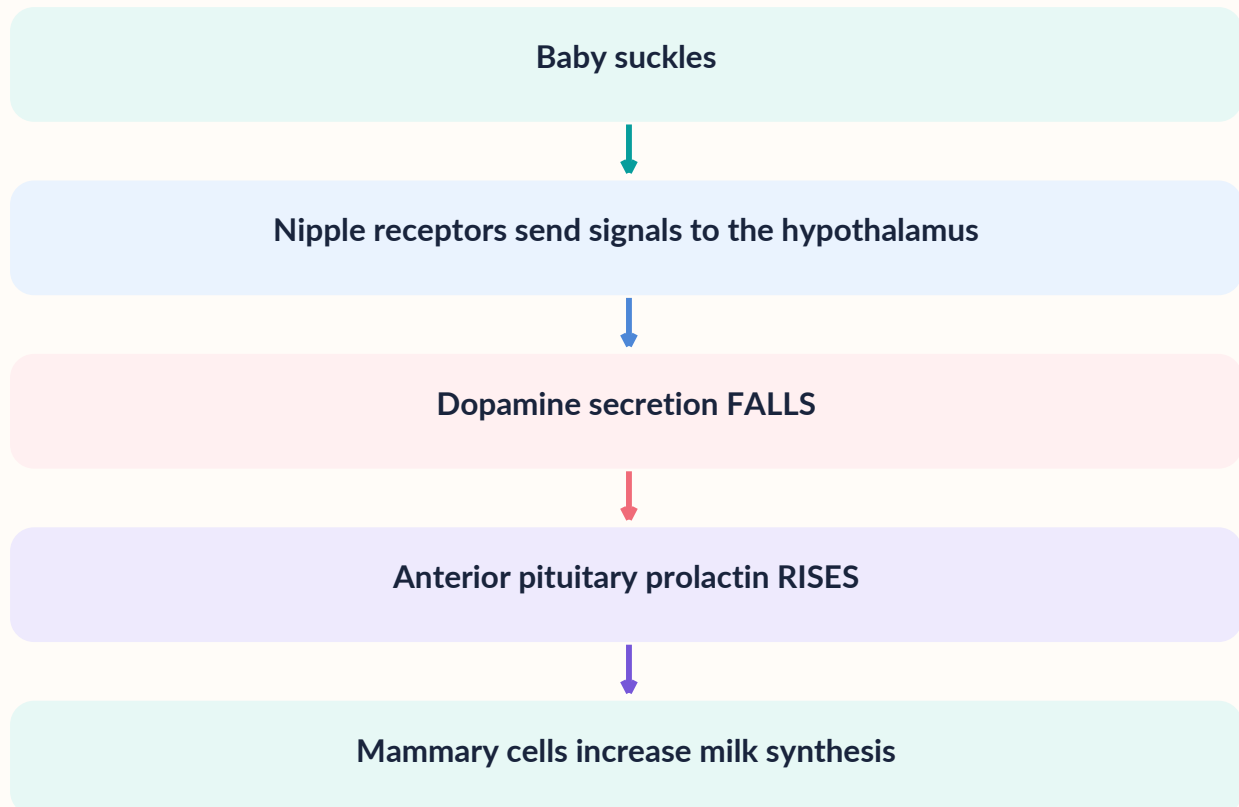
Prolactin makes milk; oxytocin ejects milk. They work as a team, but they are different hormones with different primary functions.

The suckling-to-prolactin reflex

Slide 26: use the diagram to understand disinhibition, not just memorise arrows.



Original postpartum graph + reflex flowchart: slide 26.

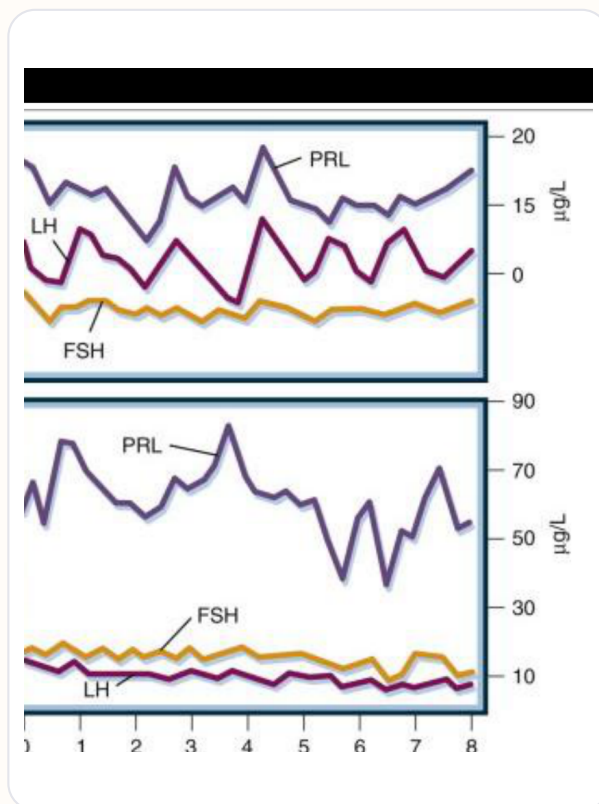


THE BRAKE PEDAL

Dopamine acts like a foot on the prolactin brake. Suckling **lifts the foot**: less dopamine -> less inhibition -> a prolactin pulse.

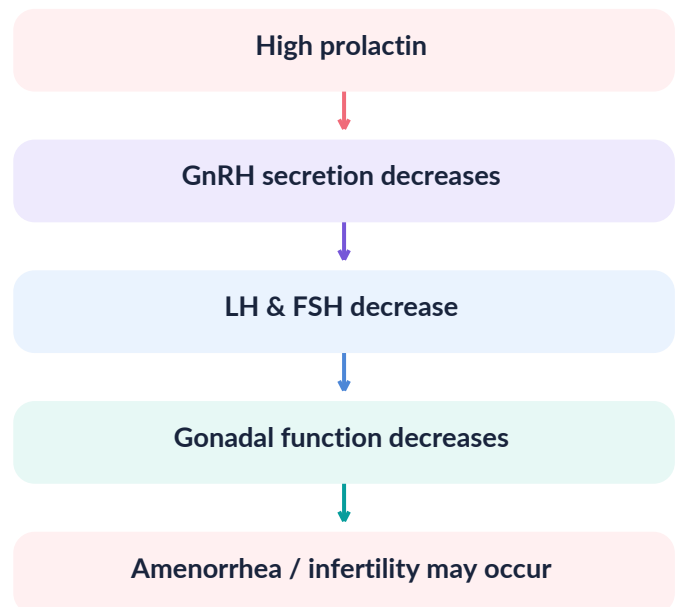
Why high prolactin can stop periods

Slides 27-28: high PRL can disturb the hypothalamic reproductive circuit.



Hormone-profile comparison: slide 27.

THE DOMINO EFFECT



CLINICAL LANGUAGE

What the terms mean

● Amenorrhea

Absence of menstruation. During breastfeeding this may be lactational amenorrhea.

● Galactorrhea

Milk discharge outside normal lactation. It can occur with high prolactin.

In men, high prolactin commonly presents with infertility or reduced libido. The lecture also links high prolactin to hypogonadism.



ONE PATHWAY EXPLAINS TWO SLIDES

Breastfeeding and a prolactinoma can both raise prolactin - but their causes differ. **PRL up -> GnRH down -> LH/FSH down.**

Three mini-cases to think through

Apply the arrows from slides 17, 23 and 26-28; explanations are on this page.

CASE A

A patient has a prolactin-producing pituitary adenoma. What changes first?

THINK IT THROUGH

Excess prolactin is secreted. High PRL can suppress GnRH -> reduce LH/FSH -> interfere with reproductive function.

CASE B

A non-secreting tumour grows upwards from the pituitary. Why ask about vision?

THINK IT THROUGH

A growing mass may compress structures near the pituitary, especially the optic chiasm, causing visual field disturbances.

CASE C

A baby suckles. Why can maternal prolactin rise without a new releasing hormone?

THINK IT THROUGH

Suckling reduces hypothalamic dopamine release. Removing tonic inhibition (disinhibition) lets lactotrophs secrete more prolactin.



EXAM SKILL

When you see a hormone problem, ask: **too much secretion, too little secretion, or a mass-effect problem?** That question separates several diagnoses in this lecture.

12 tiny facts that make it stick

Each one is a recall hook for a particular lecture slide - the last adds clearly labelled extra context.

01 SLIDE 4-5

Two delivery trucks

Magno delivers ADH/oxytocin via long axons; parvo sends regulatory hormones by portal blood.

02 SLIDE 7

Express-lane chemistry

Portal blood carries hypothalamic signals directly to anterior pituitary cells before systemic dilution.

03 SLIDE 7

A barrier with windows

Fenestrated local capillaries allow hormones to move between blood and tissue.

04 SLIDE 10

Your mouth helped build it

The anterior pituitary starts as an upgrowth from embryonic oral ectoderm.

05 SLIDE 11

A pouch can leave clues

Colloid-filled cysts in pars intermedia are remnants of Rathke's pouch.

06 SLIDE 14

Growth cells dominate

Somatotrophs form about 35-45% of anterior endocrine cells in the lecture.

07 SLIDE 14

One cell, two outputs

One gonadotroph cell type releases BOTH FSH and LH.

08 SLIDE 22

Shared alpha, unique beta

LH / FSH / TSH use a shared alpha subunit plus hormone-specific beta subunits.

09 SLIDE 22

POMC = parent molecule

A common precursor gives rise to ACTH, LPH and MSH.

10 SLIDE 17-18

Benign can still be a bully

A pituitary mass can cause problems by compressing normal tissue or the optic chiasm.

11 SLIDE 26

Milk uses a brake release

Suckling lowers dopamine inhibition and enables a prolactin pulse.

12 SLIDE 26

Milk: make vs move

Prolactin makes milk; oxytocin ejects milk. EXTRA CONTEXT beyond slide 26.



HOW TO USE THESE

Read the title of each card, hide the explanation, and reconstruct the fact in your own words.

Your 60-second exam cheat sheet

Use this as your last look before testing yourself.

ANATOMY + DEVELOPMENT

Anterior: oral ectoderm / Rathke's pouch / portal control.

Posterior: neural tissue / long axons / releases ADH + oxytocin.

Pars intermedia: remnant cysts from Rathke's pouch.

FIVE CELLS -> SIX HORMONES

Somatotroph	GH	GHRH (+); somatostatin (-)
Lactotroph	PRL	Dopamine (-)
Gonadotroph	FSH + LH	GnRH (+)
Thyrotroph	TSH	TRH (+)
Corticotroph	ACTH	CRH (+)

PROLACTIN LOGIC

Suckling: dopamine falls -> prolactin rises -> milk production.

High prolactin: GnRH falls -> LH/FSH fall -> reproductive dysfunction.



FINAL MNEMONIC

S-L-G-T-C = Soma, Lacto, Gonado, Thyro, Cortico. Ask the cell name, then its product, then the hypothalamic regulator.

Mini-quiz: close your notes

Try all 12 without looking back. Answer key is on the next page.

1. Which hypothalamic neuron type releases factors into portal blood?
2. Which lobe develops from Rathke's pouch?
3. What structure is vulnerable to compression by an upward-growing pituitary tumour?
4. Which anterior pituitary cell secretes both FSH and LH?
5. Name the two acidophil cell types.
6. Which anterior pituitary cell type is the most abundant in the lecture table?
7. What are the three hormones that share an alpha subunit?
8. What precursor gives rise to ACTH, LPH and MSH?
9. A recessive PIT1 mutation in the lecture mouse model lowers which hormones?
10. During suckling, does dopamine rise or fall?
11. Which hormone drives milk synthesis, and which drives milk ejection?
12. Complete the chain: high PRL -> low ____ -> low LH/FSH.

Answer key + memory reset

Check the mechanism in each explanation, not just the letter of an answer.

- 1 Parvocellular neurons: short axons -> portal capillaries.
- 2 Anterior pituitary (adenohypophysis); derived from oral ectoderm.
- 3 Optic chiasm (can cause visual field disturbances).
- 4 Gonadotrophs.
- 5 Somatotrophs and lactotrophs.
- 6 Somatotrophs, approximately 35-45%.
- 7 LH, FSH and TSH. Shared alpha; distinct beta subunits.
- 8 POMC: pro-opiomelanocortin.
- 9 PRL, GH and TSH.
- 10 Dopamine falls -> release of PRL inhibition.
- 11 PRL -> milk synthesis; oxytocin -> milk ejection (extra context).
- 12 GnRH. High PRL can suppress GnRH -> LH and FSH fall.

ABOUT THE FACTS & PICTURES

Core facts and embedded image excerpts follow **MDSA20030 - The Anterior Pituitary Gland**, with source slide numbers printed throughout. The "milk ejection" distinction is clearly labelled as extra context, beyond the supplied slides.

Study tip: cover the right-hand side of the hormone tables and recall each answer aloud.